

Seismic STA/LTA Detection

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1 Methodology

1.1 Overview

This study implements a three-component seismic waveform processing procedure for the detection of P-wave and S-wave arrivals. The input data consist of acceleration records measured in the East–West, North–South, and vertical directions. The three components are denoted as $AccE$, $AccN$, and $AccV$, respectively.

The workflow includes the following main steps:

1. Load the three-component acceleration records.
2. Remove baseline trends and filter the acceleration signals.
3. Integrate acceleration to obtain velocity and displacement.
4. Detect the P-wave arrival using the STA/LTA method on the vertical component.
5. Detect the S-wave arrival using strong peaks on the horizontal components.
6. Estimate the source-to-station distance from the P–S arrival-time difference.
7. Visualize the waveform, STA/LTA ratio, and time–frequency characteristics.

The procedure is designed for single-station preliminary seismic analysis. It provides approximate P-wave and S-wave arrival times and an approximate epicentral distance. The detection results should be interpreted carefully because they depend on waveform quality, noise level, filter settings, velocity assumptions, and threshold parameters.

1.2 Three-Component Acceleration Records

A seismic station generally records ground motion in three orthogonal directions. In this study, the acceleration components are defined as

$$AccE(t) : \text{East–West acceleration,} \tag{1}$$

$$AccN(t) : \text{North–South acceleration,} \tag{2}$$

and

$$AccV(t) : \text{ vertical acceleration.} \quad (3)$$

The vertical component is mainly used for P-wave detection. This choice is based on the common observation that the initial P-wave motion may be more distinguishable on the vertical channel, especially for local or regional earthquakes. The horizontal components are used for S-wave detection because S waves often produce relatively strong horizontal ground motion.

The waveform data are imported from one-column numerical files. Before subsequent analysis, the code verifies that each input file contains numeric data and converts the input waveform into a column vector. The time vector is defined by

$$t_i = \frac{i - 1}{F_s}, \quad (4)$$

where F_s is the sampling frequency and i is the sample index. In the present implementation,

$$F_s = 200 \text{ Hz}, \quad (5)$$

which means that the sampling interval is

$$\Delta t = \frac{1}{F_s} = 0.005 \text{ s.} \quad (6)$$

1.3 Signal Preprocessing

1.3.1 Detrending

Raw seismic acceleration data may contain a baseline offset or a slowly varying linear trend caused by instrument response, environmental effects, or data acquisition errors. If such trends are not removed, numerical integration can produce unrealistic velocity and displacement drift.

Therefore, each acceleration record is first linearly detrended:

$$a_d(t) = a(t) - \hat{a}_{\text{trend}}(t), \quad (7)$$

where $a(t)$ is the original acceleration signal and $\hat{a}_{\text{trend}}(t)$ is the estimated linear trend. The detrended signal $a_d(t)$ is used in later filtering and integration steps.

1.3.2 Low-Pass and High-Pass Filtering

The preprocessing function applies a low-pass filter followed by a high-pass filter. Together, these operations form an effective band-pass filtering procedure.

The low-pass filter suppresses high-frequency noise above the selected cutoff frequency:

$$f_{\text{LP}} = 20 \text{ Hz.} \quad (8)$$

The high-pass filter removes long-period baseline drift and very low-frequency noise below

$$f_{\text{HP}} = 0.8 \text{ Hz.} \quad (9)$$

A fourth-order Butterworth filter is used. The normalized cutoff frequency required by the MATLAB `butter` function is calculated relative to the Nyquist frequency:

$$f_N = \frac{F_s}{2}. \quad (10)$$

Thus, the normalized cutoff frequency is

$$W_n = \frac{f_c}{f_N}. \quad (11)$$

For the current sampling frequency, the Nyquist frequency is 100 Hz. Therefore, the normalized low-pass and high-pass frequencies are $20/100 = 0.2$ and $0.8/100 = 0.008$, respectively.

The MATLAB function `filtfilt` is used instead of the causal `filter` function. The `filtfilt` method applies the filter forward and backward in time, producing approximately zero-phase filtering. This is important for arrival-time picking because it avoids a phase delay caused by conventional causal filtering.

The filtered acceleration can be represented conceptually as

$$a_f(t) = H_{\text{HP}} [H_{\text{LP}} [a_d(t)]] , \quad (12)$$

where H_{LP} and H_{HP} denote the low-pass and high-pass filtering operations.

1.4 Velocity and Displacement Calculation

Acceleration is converted to velocity and displacement by numerical integration. The velocity is calculated from filtered acceleration as

$$v(t) = \int_0^t a_f(\tau) d\tau. \quad (13)$$

The displacement is then calculated from velocity:

$$u(t) = \int_0^t v(\tau) d\tau. \quad (14)$$

In the implementation, the trapezoidal numerical integration method is used through the MATLAB function `cumtrapz`. For discrete time samples, the integration is approximated by summing trapezoidal areas between adjacent samples.

Because integration amplifies low-frequency errors, even a small acceleration offset can produce large velocity and displacement drift. To reduce this effect, the calculated velocity is high-pass filtered before the second integration. Finally, a linear detrending operation is applied to the displacement signal.

The outputs of the preprocessing stage are therefore:

- filtered acceleration $a_f(t)$,
- drift-reduced velocity $v(t)$,
- drift-reduced displacement $u(t)$.

Although the current main procedure primarily uses the vertical component for P-wave detection, the same preprocessing procedure is applied independently to $AccE$, $AccN$, and $AccV$.

2 P-Wave Detection Using STA/LTA

2.1 Principle of STA/LTA

The short-term average to long-term average method, commonly called STA/LTA, is a standard approach for automatic seismic event detection. The basic idea is that background noise has relatively stable amplitude, while a seismic arrival causes a rapid increase in waveform energy.

For an input signal $x(i)$, the characteristic amplitude function is defined as

$$A(i) = |x(i)|. \quad (15)$$

The short-term average is calculated over a short window with N_{STA} samples:

$$\text{STA}(i) = \frac{1}{N_{\text{STA}}} \sum_{j=0}^{N_{\text{STA}}-1} A(i-j). \quad (16)$$

Similarly, the long-term average is calculated over a longer window with N_{LTA} samples:

$$\text{LTA}(i) = \frac{1}{N_{\text{LTA}}} \sum_{j=0}^{N_{\text{LTA}}-1} A(i-j). \quad (17)$$

The STA/LTA ratio is then

$$R(i) = \frac{\text{STA}(i)}{\text{LTA}(i)}. \quad (18)$$

Before the seismic arrival, the short-term and long-term amplitudes are similar, and the ratio is usually close to one. When the P wave arrives, the short-term amplitude increases more rapidly than the long-term amplitude. Consequently, the ratio increases and may exceed a predefined trigger threshold.

2.2 Limitations of STA/LTA Picking

STA/LTA is computationally efficient and appropriate for automatic preliminary detection. However, it does not always identify the exact physical first break of the P wave. The detected onset can be affected by the following factors:

- background noise level,
- local cultural or instrumental noise,
- filter frequency band,
- STA and LTA window lengths,
- trigger thresholds,
- waveform amplitude and signal-to-noise ratio.

A lower trigger threshold may detect weak arrivals earlier, but it also increases the risk of false triggers. A higher threshold is more conservative but may delay the detected arrival or miss weak events. Therefore, the STA/LTA parameters should be adjusted according to the station environment and the characteristics of the seismic dataset.

3 S-Wave Detection from Horizontal Components

3.1 Detection Strategy

After P-wave detection, the algorithm estimates the S-wave arrival from the two horizontal acceleration components, $AccE$ and $AccN$. The method assumes that the S-wave arrival is associated with a relatively strong horizontal motion after the P-wave arrival.

To avoid selecting a peak associated with the P wave, the search begins after a prescribed delay:

$$T_{\text{start}} = T_P + T_{\text{PS,min}}, \quad (19)$$

where the current minimum P–S separation is

$$T_{\text{PS,min}} = 0.5 \text{ s}. \quad (20)$$

The corresponding starting sample is

$$i_{\text{start}} = \lfloor (T_P + T_{\text{PS,min}})F_s \rfloor + 1. \quad (21)$$

For each horizontal component, the absolute waveform amplitude is used:

$$A_E(i) = |AccE(i)|, \quad (22)$$

$$A_N(i) = |AccN(i)|. \quad (23)$$

A peak is considered significant when its amplitude exceeds a fraction of the maximum amplitude in the remaining search interval:

$$A(i) \geq \alpha A_{\text{max}}, \quad (24)$$

where the threshold fraction is

$$\alpha = 0.72. \quad (25)$$

The first detected significant peak on each component provides two candidate times, $T_{S,E}$ and $T_{S,N}$. The selected S-wave arrival time is the earlier valid candidate:

$$T_S = \min(T_{S,E}, T_{S,N}). \quad (26)$$

The corresponding component is reported as either East–West or North–South.

4 Distance Estimation from the P–S Time Difference

4.1 Travel-Time Relationship

The difference between S-wave and P-wave arrival times can be used to estimate the source-to-station distance. Let D be the travel distance, V_P be the P-wave velocity, and V_S be the S-wave velocity. The arrival times can be written as

$$T_P = \frac{D}{V_P}, \quad (27)$$

and

$$T_S = \frac{D}{V_S}. \quad (28)$$

Subtracting the two expressions gives

$$T_S - T_P = D \left(\frac{1}{V_S} - \frac{1}{V_P} \right). \quad (29)$$

Solving for distance gives

$$D = (T_S - T_P) \frac{V_P V_S}{V_P - V_S}. \quad (30)$$

This equation is implemented in the distance calculation function.

4.2 Velocity Model

The current code assumes a P-wave velocity of

$$V_P = 6.0 \text{ km/s}. \quad (31)$$

The S-wave velocity is approximated using

$$V_S = \frac{V_P}{\sqrt{3}}. \quad (32)$$

Therefore,

$$V_S \approx 3.46 \text{ km/s}. \quad (33)$$

This assumption is consistent with an approximate elastic-wave relationship under simplified conditions. However, actual P-wave and S-wave velocities depend on local geology, crustal structure, focal depth, and propagation path. Thus, the calculated distance is an approximate source-to-station distance rather than a precise hypocentral or epicentral distance.

4.3 Uncertainty

The distance estimate is sensitive to errors in both P-wave and S-wave picking. If the P-wave pick has an uncertainty δT_P and the S-wave pick has an uncertainty δT_S , the uncertainty in the P–S time difference directly affects the estimated distance.

In addition, a single station cannot independently determine the full earthquake location. It can provide a distance estimate, but not a unique epicenter direction. Reliable earthquake location normally requires observations from multiple stations and a regional velocity model.